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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	NylonDaBest
Selected Domain	Domain 2 — Orbyte Smart SME Challenge: Less Busywork, Better Business (Alpha Red Solutions Sdn Bhd)
Selected SDG Track (1–17)	SDG 8 (primary); SDG 12 (secondary)
Project / Solution Name	SupplySwipe (agent name: Wakil)
Name (Leader)	Kayla Elisheva Siwi
Email & Phone (Leader)	kaykayelisheva@gmail.com / +62 812-8000-6835
University Affiliation(s)	Monash University Malaysia

Number of Team Members 4

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysia's 1.09 million MSMEs include tens of thousands of cafés, restaurants and stalls that buy fresh ingredients every week from small suppliers. Prices move constantly: Grade C eggs rose 7.3% in a single quarter of 2026 and standard chicken climbed to RM9.70/kg as farms closed. So owners renegotiate with three to eight suppliers almost every week, and they do it on WhatsApp: a price question, a counter, a delivery question, a phone call, then an invoice that does not match the chat. Finding a new supplier is worse, because every candidate means the same rounds of messages about price, quantity, delivery and payment terms. Small suppliers face the mirror image, repeating the same quote to dozens of leads and chasing confirmations. Terms agreed in scattered chats are easy to misquote: roughly 39% of manually handled invoices contain errors, and invoice discrepancies are the most common supplier complaint in comparable grocery supply chains. The cost is hours of messaging, lost deals, disputes and over-ordering that ends as waste. Large buyers have already solved this. Walmart lets AI negotiate with suppliers and 82% of those suppliers preferred it; Alibaba's Accio negotiates sourcing but requires a human to approve every price and order. None of this reaches a 40-seat café and a poultry wholesaler in Shah Alam. They have no buying power, no procurement staff and no system of record, only a phone. The burden is not ordering; it is negotiating and remembering what was agreed.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

SupplySwipe is an Orbyte module in which each business has a private AI agent, its Wakil, that negotiates supply deals while owners approve instead of chasing. Buyers post a request (item, quantity, delivery days, public target, sealed maximum); suppliers list items (list price, sealed floor, MOQ, capacity, areas, terms). A rule-based feed, with no AI, applies hard filters (halal, category, delivery area, quantity versus capacity) and ranks counterparts with "why this match" chips. A swipe right is the trigger. Both Wakils load their own sealed limits, style and memories of past dealings, plus the request, the listing and the PriceCatcher market price, and exchange at most 20 structured messages: ask, answer, offer, counter, accept or walk away. Code checks every message before it is sent: offers outside the agent's own limits, drafts that leak a sealed value, and terms not in the request or listing are rejected and replaced by a rule-based move. The language model is used only where it adds value: reading free-text specs, asking clarifying questions and writing plain-language reasons. A deal is only a proposal: both owners receive one notification and approve, counter through their Wakil, or reject; staff may approve only below an owner-set limit. Approval creates one deal record with pinned terms, opens a chat that cannot change those terms, and writes every step, including blocked drafts, to an append-only activity log. When the supplier's invoice arrives it is compared line by line with the pinned terms, and any mismatch is flagged before payment. Each Wakil then saves a short memory, so next week's renegotiation is faster. From the SDC workshops we use Replit for build and hosting, a GenAI API with structured outputs, and Supabase Postgres with row-level security. The negotiation engine is exposed as a reusable API for other Orbyte modules.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Layer	Choice	Why
Frontend	Next.js 14 + TypeScript + Tailwind; Recharts for the price chart; three.js (optional 3D agent with 2D fallback)	Phone-first screens, live negotiation replay, swipe deck
Backend	Python / FastAPI (negotiation engine, validator, feed ranker)	Rules and validation in code; LLM only for wording
Database	Supabase Postgres with row-level security; Realtime for live messages	Sealed limits readable only by their own business; append-only log
LLM	Anthropic Claude Haiku 5.5 (claude-haiku-5-5), structured JSON output only, per-message and per-negotiation token caps	Wording, clarifying questions, reasons; never prices
Hosting	Replit (app deployed from the team's own account; deployment URL in the declaration)	Compulsory sponsor platform
Data	Synthetic businesses, requests and listings; market reference modelled on DOSM PriceCatcher (CC BY 4.0)	No real personal or company data
Repo	github.com/kaylaelishevaa/supply-swipe, public, Apache-2.0	Judges can read every component

- System block diagram: Orbyte hub (simulated profiles and stock) and the web app (buyer and supplier views) → Backend API on Replit: feed ranker (code), negotiation engine + validator (two agents, ≤20 JSON messages), dual approval → deal record → activity log → Supabase Postgres (sealed limits, requests, listings, negotiations, approvals, log) and the GenAI API (structured JSON only).
- Did your project utilise any official SDC sponsor-provided GenAI API or Cloud Platform? Yes. Replit Core (sponsor-provided) builds and hosts the application; proof from our own account is in the Tech & Tools Declaration. RunPod is not used because no GPU work is needed. The GenAI API (Anthropic) is not sponsor-provided: it runs on the team's own account, is optional, and is not called by the hosted demo.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We began by asking café owners to show us their supplier chats and counting the messages it took to agree one deal. Every chat followed the same loop: price, counter, delivery, silence, call, mismatched invoice. That reframed the problem from "ordering" to "negotiating and remembering what was agreed". We rejected a chatbot, because the owner would still be haggling, just with software, and we rejected full autonomy, because owners will not let software commit their money. The design principle became bounded autonomy: owners set limits once, agents work inside them, and a person approves every deal. Supplier interviews made it two-sided: they repeat the same quote to dozens of leads and want a fast, fair, recorded process. Viability comes from putting rules where rules belong, with ranking and validation in code and the model used only for language, which keeps cost and errors low. Usability comes from swipe cards, one-tap approval and plain-language reasons, so a first-time user completes the journey unaided. For Phase 2 we deploy on Replit, test with five outlets and two suppliers on synthetic data, measure messages per agreed deal and limit breaches, and iterate the limits form.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target 8.3 (support MSME productivity, entrepreneurship, formalisation and growth). Secondary 8.2 (productivity through technological upgrading) and 12.3 (reduce food loss along supply chains: quantities negotiated against actual need, invoices checked against agreed terms).
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Today 0% of agreed supplier terms are captured in a structured record: deals are agreed in WhatsApp chats and calls, with no approval step and no system of record. About 39% of manually processed invoices contain errors (Resolve Pay via NetSuite). Context: 1,086,386 MSMEs (DOSM 2024); Grade C eggs +7.3% and chicken +3.3% to RM9.70/kg in Q1 2026 (Ministry of Economy); 73% of Malaysian SMEs report late payment (SME Media Group/ INTI, 2017). Sources: DOSM MSME statistics 2024 · Business Today, 28 Apr 2026 (egg prices) · NetSuite, invoice disputes (Resolve Pay) · SME.asia late-payment
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	(1) Owner messages and minutes from request to agreed terms per deal. (2) Share of deals captured in one structured record with pinned terms. (3) Agent limit breaches per 50 test negotiations (validator log). (4) Invoice mismatches caught before payment
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	≥90% fewer owner messages per agreed deal (baseline measured in the Phase 2 pilot → one approval); request-to-approved deal under 5 minutes; 100% of deals within owner limits with 0 breaches in 50 test negotiations; 100% of invoices compared to pinned terms before payment; 3 of 3

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ✓ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

- ✓ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

- ✓ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

- ✓ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Next.js 14.2 / React 18	code	MIT
TypeScript 5	code	Apache-2.0
Tailwind CSS 3.4	code	MIT
Recharts 3.10	code	MIT
three.js r160 (CDN, optional)	code	MIT
Plus Jakarta Sans, JetBrains Mono (Google Fonts)	fonts	SIL OFL
FastAPI, Pydantic, Supabase client libraries (backend)	code	MIT
Uvicorn, python-dotenv, httpx (backend)	code	BSD-3-Clause
Anthropic Claude Haiku 5.5	model (pre-trained, API)	Anthropic Commercial Terms
DOSM PriceCatcher (format reference)	dataset (public)	CC BY 4.0
Synthetic businesses, requests, listings, transcript	dataset (team-created)	n/a

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



Kayla Elisheva Siwi, Team Leader

Date: 25 October 2026